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SCIENTIFIC INVESTIGATIONS

Optimizing wearable single-channel electroencephalography sleep staging in a heterogeneous sleep-disordered population using transfer learning

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Study Objectives: Although various wearable electroencephalography devices have been developed, performance evaluation of the devices and their associated automated sleep stage classification models is mostly limited to healthy participants. A major barrier for applying automated wearable electroencephalography sleep staging in clinical populations is the need for large-scale data for model training. We therefore investigated transfer learning as a strategy to overcome limited data availability and optimize automated single-channel electroencephalography sleep staging in people with sleep disorders.

Methods: We acquired 52 single-channel frontopolar headband electroencephalography recordings from a heterogeneous sleep-disordered population with concurrent polysomnography (PSG). We compared 3 model training strategies: "pretraining" (ie, training on a larger dataset of 901 conventional PSGs), "training-from-scratch" (ie, training on wearable headband recordings), and "fine-tuning" (ie, training on conventional PSGs, followed by training on headband recordings). Performance was evaluated on all headband recordings using 10-fold cross-validation.

Results: Highest performance for 5-stage classification was achieved with fine-tuning (κ = .778), significantly higher than with pretraining (κ = .769) and with training-from-scratch (κ = .733). No significant differences or systematic biases were observed with clinically relevant sleep parameters derived from PSG. All sleep disorder categories showed comparable performance.

Conclusions: This study emphasizes the importance of leveraging larger available datasets through deep transfer learning to optimize performance with limited data availability. Findings indicate strong similarity in data characteristics between conventional PSG and headband recordings. Altogether, results suggest the combination of the headband, classification model, and training methodology can be viable for sleep monitoring in the heterogeneous clinical population.

Keywords: single-channel, wearable EEG, sleep staging, sleep disorders, transfer learning

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BRIEF SUMMARY

Current Knowledge/Study Rationale: Although various wearable electroencephalography devices have been developed, performance evaluation of the devices and their associated automated sleep stage classification models is still mostly limited to healthy participants. Because the need for large-scale data for model training, including concurrent gold-standard polysomnography, is a major barrier for applying wearable electroencephalography in the clinical population, we investigated transfer learning as model training strategy to overcome data availability limitations.

Study Impact: Results emphasize the importance of leveraging larger available datasets through deep transfer learning to optimize performance of wearable single-channel electroencephalography sleep staging in the sleep-disordered population (κ = .778). The relatively small differences between pretraining and fine-tuning training strategies suggested strong similarity in data characteristics between conventional polysomnography and the headband recordings.

INTRODUCTION

Polysomnography (PSG) is considered the gold standard for sleep monitoring and sleep disorder diagnostics.[1,2] After a PSG recording, 30-second epochs are manually scored according to American Academy of Sleep Medicine (AASM) guidelines as 1 of 5 sleep stages (wake/N1/N2/N3/rapid eye movement [REM]) through the visual inspection of patterns in different sensor traces: multiple electroencephalography (EEG) leads, electrooculography, and electromyography.[2] PSG is often acquired in a dedicated facility and can be cumbersome for a patient, possibly disrupting sleep in what is coined the "first-night effect."[3] Moreover, because a PSG is performed over only 1 or 2 nights, it is unsuitable to capture night-to-night sleep variability.

In recent years, less-obtrusive sleep monitoring systems have been developed, including wearable (single-channel) EEG headbands. Because there are no AASM scoring guidelines available for this limited montage, automated models are typically used for sleep stage scoring. This automated scoring can furthermore support their applicability for prolonged and large-scale monitoring. Performance of several headbands and their automated sleep stage classification models have been studied in cohorts composed of healthy participants, including the

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Dreem,[4] Sleep Profiler,[5,6] SmartSleep,[7] Zeo,[8] and ZMachine devices.[9,10] However, performance studies of wearable EEG-based sleep staging in sleep-disordered populations are thus far limited to a few studies with mixed samples that included patients with periodic limb movement disorder, obstructive sleep apnea, and insomnia.[11,12]

A major barrier for applying automated wearable EEG sleep staging in clinical populations is the need for large-scale data for model training, including concurrent gold-standard PSG. Because state-of-the-art deep learning classification models require hundreds of sleep recordings to achieve expert-level performance,[13–15] training a new model on a limited wearable EEG dataset (ie, "training-from-scratch") can lead to suboptimal performance. Potentially, the models can be pretrained ("pretraining") on other available sleep recordings, such as with frontal leads of standard EEG montages from larger PSG databases, and afterward these can be applied to the wearable EEG data. However, this method is susceptible to potential mismatches between the datasets, including population (eg, age and sleep disorders) or channel (eg, electrode location and electrode type) differences.[16,17] Transfer learning has been suggested as a solution to overcome both data availability limitations and data mismatches.[18,19] Here, the model would be first pretrained on the larger database, followed by some additional training (ie, "fine-tuning") on the wearable EEG data.

We investigate transfer learning as a model training strategy to overcome limited data availability and optimize performance of a publicly available automated sleep stager (TinySleepNet)[20] on data acquired using the wearable single-channel EEG SmartSleep Deep Sleep headband (Philips Respironics Germany GmbH, Herrsching am Ammersee, Germany). The performance of 3 different strategies to train the automated classification model (pretraining, training-from-scratch, and fine-tuning) was compared in a heterogeneous sleep-disordered population.

METHODS

Data

The wearable EEG recordings were obtained from the Prolonged In-home Sleep Monitoring (PRISM) study. The PRISM dataset includes both in-laboratory and at-home sleep recordings of patients who were referred to the Kempenhaeghe Center for Sleep Medicine (Heeze, The Netherlands) and were scheduled for an overnight PSG as part of standard clinical practice. The PRISM study targeted a heterogeneous sleep-disordered adult population as typically encountered in a third-line expert center for sleep, with the exception of patients actively using positive airway pressure devices. PRISM participants were first monitored for a maximum of 7 nights at home using a wearable EEG headband, a photoplethysmography wristband, and a ballistocardiography mattress band. Afterward, for 1 night these sensors were added to the concurrent in-laboratory clinical (video) PSG. The present study made use of these in-laboratory recordings.

Additional PSGs were obtained from the HealthBed dataset as well as from the Sleep and Obstructive Sleep Apnea Measuring with Non-Invasive Applications (SOMNIA) cohort, both performed at the Kempenhaeghe Center for Sleep Medicine and recorded before January 2021.[21,22] The HealthBed dataset contains healthy adults voluntarily recruited for an overnight PSG, and the SOMNIA dataset was acquired among a heterogeneous sleep-disordered population scheduled for a PSG for clinical purposes.

For all datasets, a pool of trained clinicians manually scored the PSG recordings according to the AASM scoring guidelines.[23] Primary sleep diagnoses were coded according to the *International Classification of Sleep Disorders*, third edition (ICSD-3) criteria. The PRISM, SOMNIA, and HealthBed studies were reviewed by the Maxima Medical Center medical ethical committee (Veldhoven, The Netherlands, reported under W21.068, W17.128, and N16.074, respectively) and adhere to the Declaration of Helsinki. Approval of the data analysis protocol was granted by the Philips Research Internal Committee for Biomedical Experiments and the medical ethical committee of Kempenhaeghe Center for Sleep Medicine. Informed consent was provided by all participants before participation.

Classification model

We used the TinySleepNet deep learning model[20] for 5-stage (wake/N1/N2/N3/REM) automated sleep staging. TinySleepNet is a computationally efficient version of DeepSleepNet.[24] The model has demonstrated performance comparable to human interrater agreement in manual scoring and to other single-channel EEG deep learning models.[20] Its architecture comprises 4 convolutional neural networks layers interspersed with 2 max-pooling and drop-out layers, followed by a unidirectional long short-term memory layer and another drop-out layer. The model uses as input (raw) single-channel EEG data and is purely causal, allowing for real-time implementations.

Wearable EEG

We used the wearable SmartSleep Deep Sleep Headband (Philips Respironics Germany GmbH, Herrsching am Ammersee, Germany) for single-channel EEG signal acquisition. The headband is a flexible, soft fabric, integrated closed-loop audio stimulation device containing a frontopolar (Fpz) dry single-channel EEG electrode and a wet reference electrode on the right mastoid (M2). The adjustable headband holds the frontopolar electrode, a snap connector for the M2 reference, and a speaker. The headband was originally developed to target improved sleep quality through auditory stimulation during deep sleep, but in this study only the EEG recording functionality was used and no deep stimulation was applied. Hence, the wearable EEG headband currently has a deep learning model embedded that is optimized for the detection of N3 sleep, reported 95% N3 specificity in healthy, short sleepers (< 6 hours per night) through 4-stage classification (wake/combined N1+N2/N3/REM) using 6-second epochs.[25,26] Furthermore, a Cohen's kappa coefficient of agreement (κ)[27] of .73 has previously been reported for 5-stage classification in recordings of healthy sleepers acquired using the headband.[7] The performance of the currently embedded model was evaluated for comparison, but due to the better expected performance and alignment with clinical scoring guidelines (4-class and

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6-second epochs vs 5-class and 30-second epochs), training strategies were applied to the newly trained TinySleepNet model. To obtain 30-second epoch performance of the embedded model, 6-second epochs were converted to 30-second epochs by taking the most common sleep stage within 5 consecutive epochs. If no majority class was available, the classification of the third 6-epoch epoch was selected, corresponding to the center of the 30-second timeframe.

PSG

Concurrent PSG was acquired using a Greal 4K PSG recording system (Compumedics, Victoria, Australia) and PSG sensors were set up according to AASM guidelines.[2] Limb movements, respiratory events, and snoring were automatically scored by dedicated software (ProFusion PSG V.4.00, build 423; Compumedics, Victoria, Australia), and reviewed by a certified clinician. Sleep stages and other events were manually scored.

The performance of the wearable EEG automated scoring was compared to the 5-stage manual sleep stage scoring of the PSG. For model training, the F3 and F4 electrode EEG channels of the PSGs were averaged to obtain a surrogate estimate of measurements at the Fz location, which was referenced to M2. The wearable EEG and PSG data were synchronized through visual inspection and corrected for clock offset and drift. For both signals, a fifth-order Butterworth bandpass filter between 0.2 Hz and 49 Hz and a 50-Hz notch filter were applied to reduce mains interference, followed by down-sampling from 512 Hz (PSG) and 250 Hz (wearable EEG) to a common sampling frequency of 100 Hz.

Training and validation strategies

We tested 3 different training strategies: pretraining, training-from-scratch, and fine-tuning. In pretraining, the model was solely trained on the Fz-M2 (approximated using an F3 and F4 average) derivation of all available SOMNIA and HealthBed data. In training-from-scratch, the model was solely trained on the wearable (Fpz-M2) headband recordings of the PRISM dataset (only on recordings with a concurrent PSG). In fine-tuning, the model was first trained on the SOMNIA and HealthBed data, followed by additional training on the wearable headband data.

For all training strategies, we used the manually scored 5-class sleep stages based on PSG as reference. In the case of the training-from-scratch and the fine-tuning strategies, this was based on the concurrent PSG recorded alongside the wearable EEG.

For the pretraining strategy, we evaluated model performance on all available wearable recordings. For the training-from-scratch strategy and for the additional training with wearable EEG in the fine-tuning strategy, we used a 10-fold cross-validation to train and evaluate the model performance on all available headband recordings. On each iteration of the cross-validation, 80% of the recordings were used as training, 10% as validation, and 10% as test set. On each cross-validation iteration the model was trained for a maximum of 200 training iterations, with early stopping if no improvement in the weighted F1 score was observed in the validation set within the following 50 training iterations. The trained model was finally evaluated on the test set of each cross-validation iteration. Hence, the same participant was never included in more than 1 set at the same time.

For the fine-tuning strategy, weights of the pretrained model were loaded, without freezing any convolutional neural networks layers (ie, making layers nontrainable). Furthermore, the learning rate was lowered from $1e^{-4}$ to $1e^{-5}$. The settings for the fine-tuning strategy were derived from previous work with the same deep learning model for the purpose of tuning the model to different sleep disorders and electrode locations.[16]

Evaluation statistics

Performance was assessed in accordance with recent recommendations for evaluating new sleep monitoring technologies.[28] Sleep stage classification performance was analyzed using the agreement between the classification of the automated wearable EEG scoring vs manual PSG scoring on an epoch-per-epoch basis, for all stages (wake/N1/N2/N3/REM) and separately for each recording. Overall performance across classes was calculated using the Cohen's kappa coefficient of agreement (κ), weighted F1 score (harmonic mean between the positive predictive value and sensitivity), and accuracy on a per-participant basis. Sleep stage specific performance was calculated using the per-class Kappa, F1 score, sensitivity, and specificity on a per-participant basis.

Performance differences between the training strategies (pretraining, training-from-scratch, and fine-tuning) were evaluated using a within-participant design. Considering the larger sample size ($n = 52$) and the equal variance between groups (Levene's test, not further reported), parametric t tests were performed.

Sleep parameters derived from manual scoring were compared to the sleep parameters derived from the 3 trained models (pretraining, training-from-scratch, and fine-tuning) using t tests and Pearson correlation (r), due to large sample sizes and equal variance. To correct for multiple testing, a conservative Bonferroni correction of $\alpha/5$ was applied to the significance threshold, corresponding to the parameters derived from 5 sleep stages.

To test whether there were differences in performance for subgroups of patients with different sleep disorders, we divided all participants between the major ICSD-3 categories (ie, between "insomnia disorders," "sleep-related breathing disorders," "sleep-related movement disorders," "central disorders of hypersomnolence," "circadian rhythm sleep–wake disorders," and "parasomnias," which due to their distinct characteristics were split between "REM parasomnias" and "non-REM parasomnias"). Unspecified parasomnias were included in the "non-REM parasomnia" category. Patients without a primary diagnosis of sleep disorder were grouped in a "No primary diagnosis or healthy" category. When diagnoses from multiple subgroups were present for 1 patient, the patient was included in multiple subgroups. Differences between subgroups were calculated by comparing the performance of participants within 1 subgroup to all participants from the other subgroups. Due to smaller sample sizes ($n < 30$), nonparametric between-participant

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Mann–Whitney *U* tests were performed. A conservative Bonferroni correction of $\alpha/5$ was applied for multiple testing corresponding with the parameters derived from 5 subgroups that were analyzed.

RESULTS

An overview of the demographic information and the sleep disorder categories present in the combined SOMNIA and HealthBed datasets as well as in the PRISM dataset can be found in **Table 1**.

Figure 1 illustrates the sleep stage performance for each of the 3 training methods (pretraining, training-from-scratch, fine-tuning) on both group level and individual level. For 5-stage sleep classification, fine-tuning (κ = .778 ± .131) performed significantly better than pretraining [κ = .769 ± .136, $t(51)$ = 2.35, P = .023, d = .067]. Training-from-scratch (.733 ± .140) showed significantly lower performance than both pretraining [$t(51)$ = 5.49, P < .001, d = .265] and fine-tuning [$t(51)$ = 6.60, P < .001, d = .336]. Performance differences between pretraining and fine-tuning were limited, with a κ difference lower than .05 for 48 out of the 52 participants, with slightly higher performance using the fine-tuning strategy in most participants. For 5 participants, a κ < .60 was reported in at least 1 of the training strategies, and 3 participants exhibited κ < .50 in all training strategies. No visible trend was observed in the underperforming participants with respect to the sleep disorder category, stage-specific performance, or signal quality. Notably, 2 participants with lower performance belonged to the oldest age cohort (73 and 80 years).

Since the fine-tuned model achieved the highest performance, its performance is further described. **Table 2** describes the model performance for different classification tasks, with a mean of κ = .778 overall agreement with manual scoring from PSG. Model performance for different classification tasks of the pretraining and training-from-scratch strategies can be found in **Table S1** and **Table S2** in the supplemental material. Agreement was highest for the classification of wake (κ = .840) and lowest for N1 (κ = .552). For each of the sleep stages, the mean specificity was always higher than 90% and the mean sensitivity was higher than 80%, except for N1 classification, with a sensitivity of 59%. **Figure 2** illustrates the confusion matrix, obtained after pooling the classification for all epochs of all recordings (when they were part of the test set during cross-validation). The confusion matrices for all training strategies can be found in **Figure S3**, in the supplemental material including the 4-stage classification performance of the currently embedded model, which showed a κ = .615 overall agreement with manual scoring.

Table 3 describes the sleep parameters obtained from both manual scoring and automated scoring with the best-performing model. Mean comparison showed no significant difference between manual and automated scoring for any of the sleep parameters, whereas strong correlations (r >.79) were observed for all sleep parameters. **Figure 3** illustrates the Bland–Altman plots, revealing negligible systematic biases, and no evidence of proportional bias in the estimation of sleep parameters between manual scoring from PSG and automated scoring from wearable EEG. Although there was an even dispersion around the mean for the time spent in REM sleep, a relatively large number of participants (n = 5) showed underestimation using automated scoring. Within this group of 5 participants there were no observed trends in patient characteristics, nor did any participant overlap with those showing notably lower sleep staging performance in **Figure 1**.

Table 4 shows the performance of the fine-tuned 5-stage classification model in each sleep disorder category. Although

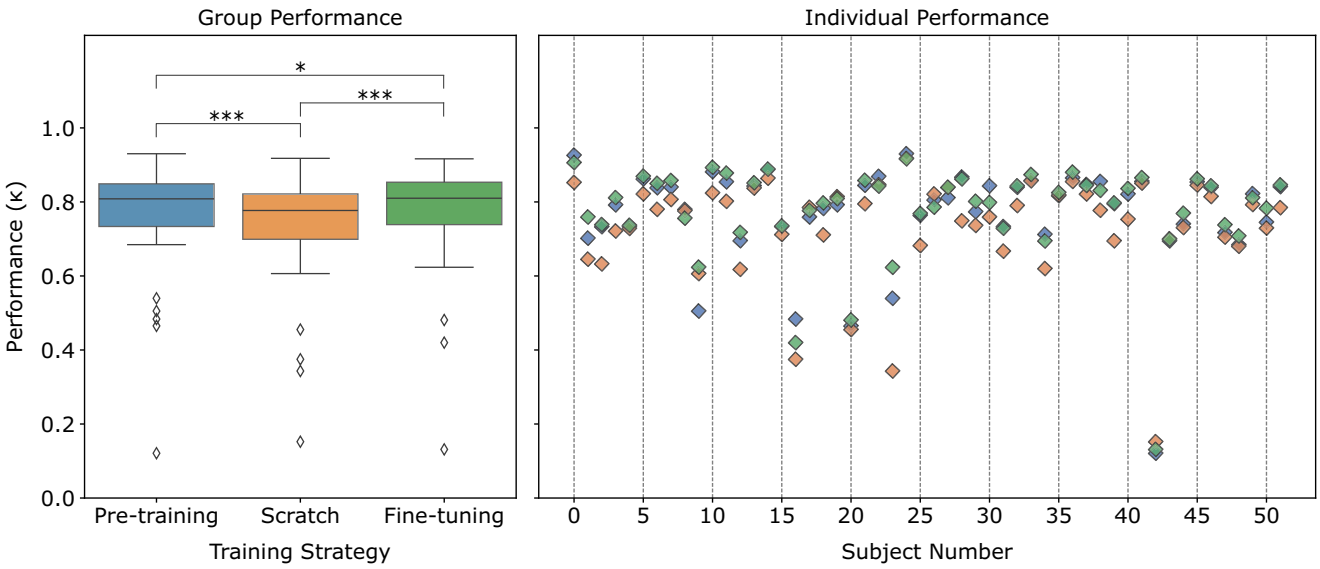
Table 1—Demographic and sleep disorder information of the combined SOMNIA and HealthBed datasets and the PRISM dataset.

	PRISM	SOMNIA + HealthBed
Participants	52	901
Age	46.3 ± 16.1 (22, 80)	46.8 ± 15.1 (18, 94)
Sex (male/female)	25/27	504/397
AHI	7.9 [4.6–20.7] (0.1, 58.2)	13.3 [7.4–26.1] (0, 91.3)
BMI (kg/m^2)	24.9 [22.7–26.6] (18.2, 37.0)	25.8 [23.8–29.3] (16.8, 48.9)
Insomnia disorders	28/52	273/901
Sleep-related movement disorders	16/52	135/901
Sleep-disordered breathing disorders	16/52	395/901
Non-REM parasomnias	8/52	69/901
REM parasomnias	4/52	64/901
Circadian rhythm sleep disorders	2/52	23/901
Hypersomnolence disorders	1/52	48/901
No primary diagnosis or healthy	1/52	134/901

The mean ± standard deviation, or the median [interquartile range] is shown, followed by the range in parentheses. Sleep disorders are grouped according to the major categories of the *International Classification of Sleep Disorders*, third edition with a further split between REM and non-REM parasomnias. AHI = apnea-hypopnea index, BMI = body mass index, PRISM = Prolonged In-home Sleep Monitoring, REM = rapid eye movement, SOMNIA = Sleep and Obstructive Sleep Apnea Measuring with Non-Invasive Applications.

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Figure 1—Group performance (left) and individual performance (right) per training strategy.



Pretraining in blue, training-from-scratch in orange, and fine-tuning in green. *P < .05, ***P < .001.

agreement seemed lower in sleep-disordered breathing (median [interquartile range]: .776 [.715–.835]) and REM parasomnia (median [interquartile range]: .754 [.645–.816]), differences with other sleep disorders were not significant.

DISCUSSION

This study aimed to optimize wearable single-channel EEG automated sleep staging performance through transfer learning to overcome limited data availability for model training. High agreement for 5-stage automated classification in data obtained with an EEG headband was reported when compared to the manual scoring of concurrent clinical PSG in a heterogeneous patient population. Highest performance was reached when the model was first trained on conventional EEG electrodes in a large database of healthy and sleep-disordered patients and afterward fine-tuned on the wearable EEG data (κ = .778 ± .131 for 5-stage classification). Performance was slightly higher compared to directly applying the pretrained model on the wearable EEG data (κ = .769 ± .136), with only small differences in performance on individual patients, suggesting strong similarity in data characteristics between conventional PSG and the wearable EEG signal. Hence, limited data mismatches were introduced by a slight change in electrode location (Fz vs Fpz) and electrode type (wet vs dry). These results are in line with earlier reporting that dry electrodes can obtain similar performance and signal quality, in general and for sleep staging in particular.[29,30]

Both pretraining and fine-tuning strongly outperformed training-from-scratch (κ = .733 ± .140), emphasizing the importance of leveraging larger datasets for model training in data-limited scenarios, such as for the validation of new monitoring technologies in clinical populations. No statistical comparison

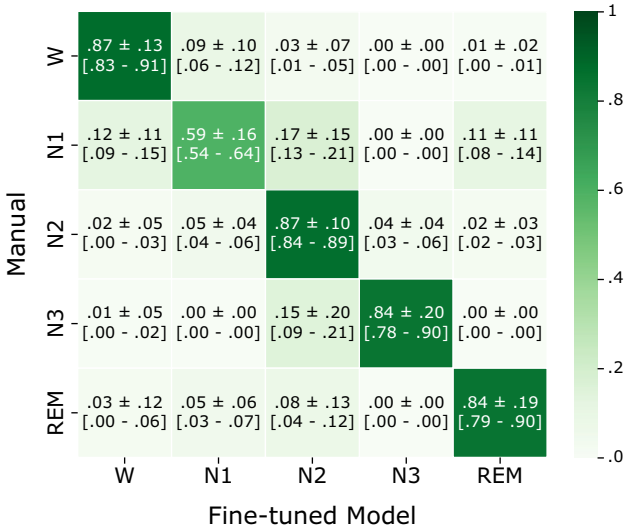
Table 2—Overall and sleep stage–specific performance for the fine-tuned classification model.

	Kappa	F1 Score	Sensitivity	Specificity
Overall	.778 ± .131 [.741–.815]	84.4 ± 9.3 [81.8–87.0]	n/a	n/a
Wake	.840 ± .143 [.800–.880]	87.2 ± 11.0 [84.1–90.3]	87.2 ± 12.7 [83.4–90.9]	97.0 ± 5.1 [95.5–98.4]
N1	.552 ± .159 [.478–.567]	56.6 ± 15.1 [52.4–60.9]	59.0 ± 17.3 [54.4–63.6]	95.2 ± 3.6 [94.2–96.2]
N2	.774 ± .147 [.733–.816]	86.6 ± 9.2 [84.0–89.2]	86.7 ± 9.6 [84.0–89.4]	91.2 ± 7.7 [89.0–93.4]
N3	.796 ± .178 [.746–.846]	81.7 ± 17.5 [76.8–86.8]	84.2 ± 20.3 [78.4–89.9]	97.8 ± 2.0 [97.2–98.4]
REM	.792 ± .119 [.735–.848]	83.2 ± 15.6 [78.7–87.7]	84.5 ± 19.3 [78.9–90.0]	97.5 ± 2.3 [96.9–98.1]

Performance calculated on a per-participant basis, reported as mean ± standard deviation [95% confidence interval]. For the overall F1 score, the weighted F1 score is reported. n/a = not applicable, REM = rapid eye movement.

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Figure 2—Confusion matrix between manual scoring and the fine-tuned 5-stage classification model.



Calculated on a per-participant basis, reported as mean ± standard deviation [95% confidence interval]. Darker hue indicates higher accuracy. REM = rapid eye movement, W = wake.

was made between the newly trained classification models and the currently embedded model because the embedded model was optimized for high N3 specificity and conversion from 6- to 30-second epochs was required.[25,26] Nevertheless, the high N3 specificity of the embedded model was confirmed in the current study and can be observed in the corresponding confusion matrix (see **Figure S3**).

This is the first wearable EEG sleep staging study conducted in a heterogeneous sleep-disordered population as encountered in a sleep clinic, including a wide age range. Results suggest that automated single-channel EEG sleep staging could be more challenging in sleep-disordered breathing and REM parasomnias, albeit this specific comparison was limited by small sample sizes and differences in performance between these subgroups were not significant. Notably, the largest constituents of sleep-disordered breathing and REM parasomnias categories in this study, namely obstructive sleep apnea and REM sleep behavior disorder, have previously been associated with lower automated sleep staging performance when compared to healthy participants and those with insomnia disorders.[16] The lower performance could potentially (partly) be attributed to the decreased sleep stage stability in these disorders, and REM sleep behavior disorder also contains distinctive pathophysiological characteristics that an automated model needs to learn.[16,31,32]

Performance in the current study (κ = .78) was higher than that reported by other headband studies in healthy participants[4,6–10] and in mixed populations including patients with periodic limb movement disorder, obstructive sleep apnea, or insomnia,[11,12] which reported kappa scores between .57 and .75 for 5-stage classification. Given the estimated human interrater scoring agreement using full PSG, which ranges between κ = .71 and .81 with an average of κ = .76,[33,34] and the increased difficulty of scoring in sleep-disrupting disorders,[35] the automated scoring performance reported here falls within this range while using a more limited montage (ie, no electromyography, electrooculography, central EEG, and occipital EEG). Although it remains unknown to what extent performance can be attributed to the headband signal quality, the model, and the model training, the improved performance using the pretraining and fine-tuning strategies emphasizes the importance of using larger datasets through transfer learning as a tool for model optimization. We have applied the most common transfer learning method for the fine-tuning strategy, involving continued learning to tune to the new dataset, while lowering the learning rate, possibly combined with making some layers nontrainable (not used in the current study). Similar approaches have been used previously to tune automated sleep staging models to different sensor modalities (ie, from EEG to electrooculography),[17] different channel locations (ie, from C4-A1 to Fpz-Cz and behind-the-ear EEG),[19] specific sleep disorders (ie, insomnia, obstructive sleep apnea,

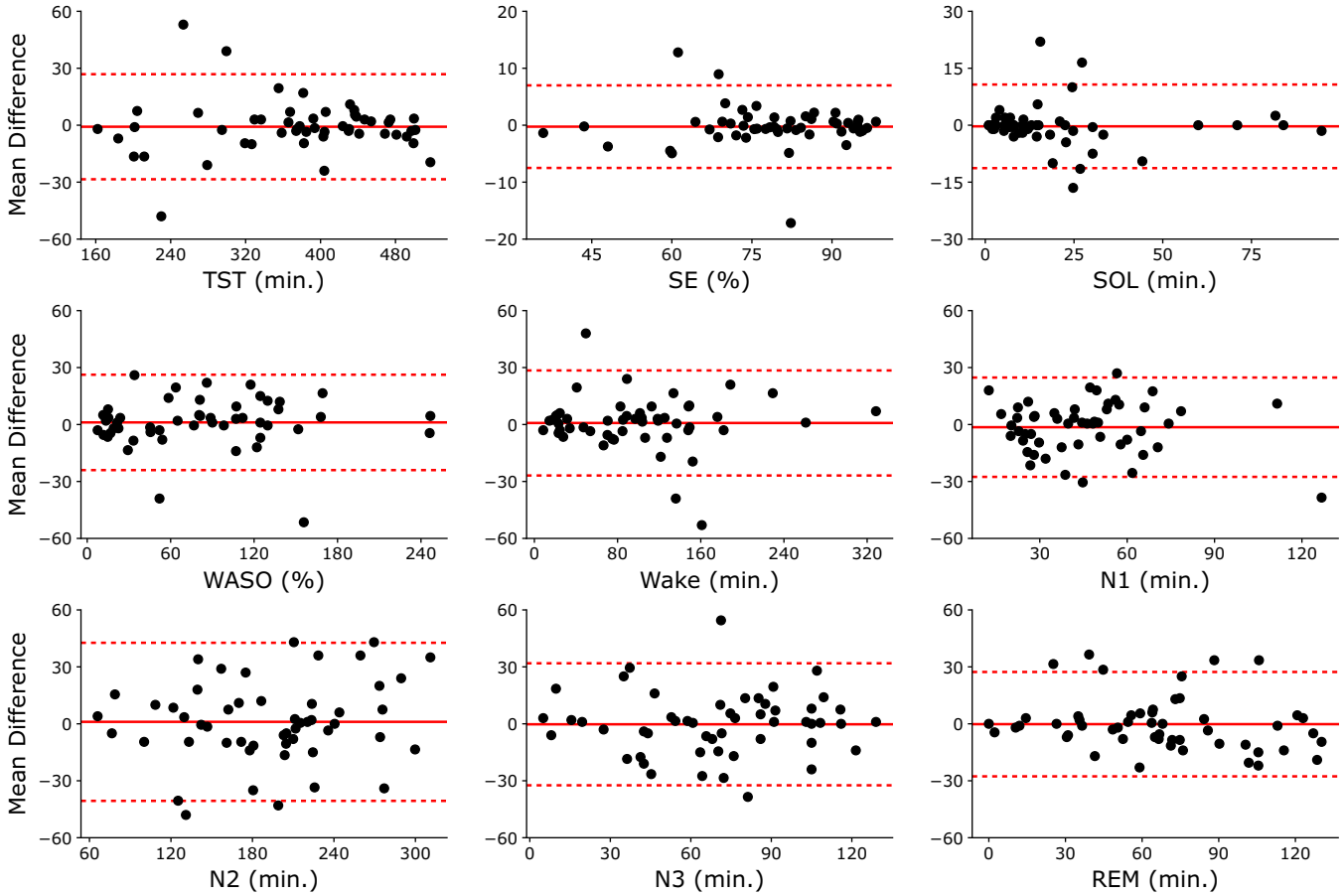
Table 3—Sleep parameters derived from manual scoring and automated scoring using the fine-tuned 5-stage classification model.

	Manual Scoring	Automated Scoring	Difference	Statistic	Correlation
TST (minutes)	374.6 ± 93.9	375.4 ± 93.7	0.8 ± 14.1	T = –.41, P = .68	r = .99, P < .001*
SE (%)	78.8 ± 13.8	79.2 ± 13.9	0.3 ± 3.7	T = –.49, P = .62	r = .96, P < .001*
SOL (minutes)	19.4 ± 21.7	19.7 ± 22.1	0.3 ± 5.6	T = –.40, P = .70	r = .92, P < .001*
WASO (%)	80.4 ± 58.9	79.3 ± 58.6	−1.1 ± 12.8	T = .62, P = .54	r = .98, P < .001*
Wake (minutes)	99.8 ± 66.0	99.0 ± 66.6	−0.8 ± 14.1	T = .41, P = .68	r = .97, P < .001*
N1 (minutes)	44.8 ± 22.4	46.2 ± 23.6	1.4 ± 13.3	T = –.76, P = .45	r = .79, P < .001*
N2 (minutes)	193.5 ± 61.7	192.4 ± 58.5	−1.0 ± 21.2	T = .35, P = .73	r = .94, P < .001*
N3 (minutes)	69.8 ± 32.5	70.1 ± 32.2	0.2 ± 16.4	T = –.11, P = .91	r = .87, P < .001*
REM (minutes)	66.6 ± 33.5	66.7 ± 36.1	0.2 ± 14.0	T = –.09, P = .93	r = .93, P < .001*

Mean and standard deviation are shown, followed by the difference between methods (automated scoring minus manual scoring), mean comparison, and correlation analysis. The asterisks indicate statistical significance. REM = rapid eye movement, SE = sleep efficiency, SOL = sleep onset latency, TST = total sleeping time, WASO = wake after sleep onset.

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Figure 3—Bland–Altman plot for each sleep parameter.



Computed by subtracting the mean of automated classification from manual classification. The solid red line indicates mean difference, and dotted red lines indicate the 95% confidence interval. REM = rapid eye movement, SE = sleep efficiency, SOL = sleep onset latency, TST = total sleep time, WASO = wake after sleep onset.

Table 4—Performance (kappa, F1 score, and accuracy) for each sleep disorder category using the fine-tuned 5-stage classification model.

	n	Kappa	Statistic	F1 Score	Statistic	Accuracy	Statistic
Insomnia disorder	28	.828 [.759–.858]	*U* = 375.0, *P* = .47	87.8 [83.8–90.3]	*U* = 375.0, *P* = .48	87.8 [83.9–90.2]	*U* = 371.0, *P* = .53
Sleep-related movement disorder	16	.807 [.761–.851]	*U* = 275.0, *P* = .80	86.9 [83.4–89.4]	*U* = 285.0, *P* = .96	86.9 [83.7–89.1]	*U* = 278.0, *P* = .85
Sleep-disordered breathing	16	.776 [.715–.835]	*U* = 203.0, *P* = .09	84.0 [79.4–88.9]	*U* = 201.0, *P* = .12	84.0 [79.9–88.4]	*U* = 212.0, *P* = .13
Non-REM parasomnia	8	.827 [.791–.853]	*U* = 197.0, *P* = .61	87.2 [87.2–89.3]	*U* = 186.0, *P* = .81	87.6 [84.9–89.2]	*U* = 190.0, *P* = .74
REM parasomnia	4	.754 [.645–.816]	*U* = 57.0, *P* = .19	81.7 [73.2–86.6]	*U* = 56.0, *P* = .18	82.7 [74.9–86.5]	*U* = 57.0, *P* = .19

Median values are shown with interquartile range in brackets, followed by statistical comparison. Statistical comparisons were performed for each subgroup, against the participants that did not have the given sleep disorder. Note that some participants had multiple primary sleep disorders. No significant differences were found, which were otherwise indicated with an asterisk. The F1 score (harmonic mean between sensitivity and positive predictive value) was weighted according to class distribution. REM = rapid eye movement.

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and REM sleep behavior disorder),[16] and to individual patients with REM sleep behavior disorder.[36] Even though other transfer learning approaches are available, this fine-tuning method is suggested to be the most robust for wearable EEG data, although a feature-matching transfer learning technique could be considered when data availability is even more restricted (ie, < 10 available recordings).[37,38]

No significant differences were observed for clinically relevant overall sleep parameters including the total sleep time, sleep efficiency, sleep onset latency, wake after sleep onset, and time spent in each sleep stage when derived from manual scoring of PSG compared to the fine-tuned model with wearable EEG. Furthermore, strong correlations in sleep parameters were observed ($r > .87$), which were only slightly lower for time spent in N1 ($r = .79$), consistent with lower interrater agreement observed in manual scoring of N1 sleep[34] and the automated detection of N1 sleep in other wearable EEG studies.[11,12] Although no systematic biases in mean differences were observed, 5 participants showed substantial underestimation of time in REM sleep. A possible explanation includes the inability of the unidirectional model for backward-looking REM classification, whereas in manual scoring AASM guidelines prescribe that "sleep preceding and contiguous with definitive REM sleep" can be scored as REM.[2,20] Furthermore, the lack of electromyography channel information potentially contributes to the underestimation, albeit frontal EEG is hypothesized to capture muscle activity to some extent.[16]

Some limitations of this study should be considered. First, although transfer learning can overcome data availability limitations for model training, analyses were still restricted by the sample size. Sleep disorder specific groups were too small for robust analysis, and the limited sample size did not allow for age-related analyses and analysis on which groups benefitted most from fine-tuning. Second, data was acquired in an in-laboratory setting and the headband was applied by a sleep technician. Hence, it remains unknown whether similar data quality would be obtained when the headband is self-applied and worn in the natural home environment. Monitoring in uncontrolled settings may benefit from the implementation of automated artifact and signal quality detection.[39] Nevertheless, the used combination of the single-channel wearable EEG headband and automated sleep classification model can be viable for prolonged at-home monitoring in a heterogeneous patient population.

Applying transfer learning techniques has shown to be beneficial for the development and model training of new wearable EEG devices. By requiring fewer training data, cumbersome and costly data collection is reduced, which is particularly advantageous in the sleep-disordered population, where minimal data acquisition is ethically preferred. How transfer learning is successfully applied is specific to the sleep stage classification model and necessitates prior experimentation to determine optimal parameter settings.[19] However, once model-specific optimal settings are established, the same approach can be repeated to tune the model to other electrode locations,[16,17] and we hypothesize comparable effectiveness when applied to different wearable single-channel EEG headbands.

In summary, results emphasize the importance of leveraging larger available datasets through deep transfer learning to overcome limited data availability and optimize automated sleep staging performance with single-channel EEG data acquired using a wearable headband. The relatively small differences between pretraining and fine-tuning strategies suggested strong similarity in data characteristics between conventional PSG and the wearable headband recordings.

ABBREVIATIONS

AASM, American Academy of Sleep Medicine
EEG, electroencephalography
ICSD, *International Classification of Sleep Disorders*
PRISM, Prolonged In-home Sleep Monitoring
PSG, polysomnography
REM, rapid eye movement
SOMNIA, Sleep and Obstructive Sleep Apnea Measuring with Non-Invasive Applications

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DISCLOSURE STATEMENT

All authors have seen and approved the manuscript. This work was performed within the IMPULS framework of the Eindhoven MedTech Innovation Center (e/MTIC, incorporating Eindhoven University of Technology, Philips Research, and Sleep Medicine Center Kempenhaeghe), including a PPS supplement from the Dutch Ministry of Economic Affairs and Climate Policy. At the time of writing, J.F.v.d.A., D.A.v.d.E., and P.F. were employed and/or affiliated with Royal Philips, a commercial company and manufacturer of consumer and medical electronic devices, commercializing products in the area of sleep diagnostics and sleep therapy. Philips had no role in the study design, decision to publish, or preparation of the manuscript. S.O. participated in advisory boards for Takeda, Jazz Pharmaceuticals, Bioprojet, and Abbvie, all paid to the institution and all unrelated to the present work. The other authors report no conflicts of interest.

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